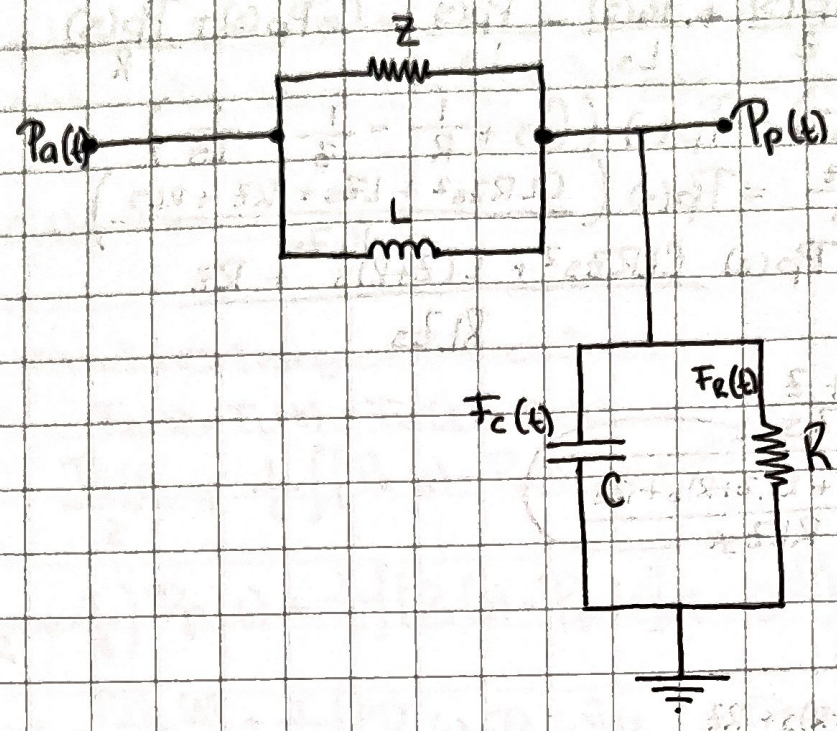
*Actualizado

/ /

5.4 Práctica: Modelado y control del sistema cardiovascular



Función de transferencia

$$F_a(t) = F_z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de LaPlace

$$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(s) = Cs\,P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$\frac{P_a(s) - P_p(s)}{z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

KYMA

$$\frac{Pa(s)}{z} - \frac{Pp(s)}{z} + \frac{Pa(s)}{Ls} - \frac{Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

$$Pa(s)\left(\frac{1}{z} + \frac{1}{Ls}\right) = Pp(s)\left(Cs + \frac{1}{R} - \frac{1}{z} - \frac{1}{Ls}\right)$$

$$Pa(s)\,\frac{Ls+z}{Lzs} = Pp(s)\left(\frac{CLRzs^2 + Lzs + Rz + RLs}{RLzs}\right)$$

$$Pa(s)\,\frac{Ls+z}{Lzs} = Pp(s)\,\frac{CLRzs^2 + L(z+R)s + Rz}{RLzs}$$

$$\frac{Pp(s)}{Pa(s)} = \frac{\frac{Ls+z}{Lzs}}{\frac{CLRzs^2 + L(z+R)s + Rz}{RLzs}}$$

Resultado

$$\frac{Pp(s)}{Pa(s)} = \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz}$$

Estabilidad en lazo abierto

→ Hipotenso

$z = 0.010$
$L = 0.001$
$R = 1.000$
$c = 3.500$

$CLRzs^2 + L(z+R)s + Rz = 0$

$a = CLRz = 3.5\times10^{-5}$
$b = L(z+R) = 1.01\times10^{-3}$
$C = Rz = 0.01$

$\lambda_1 = -14.42 + 8.8051i$
$\lambda_2 = -14.42 - 8.8051i$

Por lo tanto, es sistema presenta una respuesta estable y subamortiguada a un escalón unitario de entrada.

→ Normotenso

$z = 0.010$
$L = 0.001$
$R = 1.000$
$c = 1.500$

$a = 1.5\times10^{-5}$
$b = 1.01\times10^{-3}$
$c = 0.01$

$\lambda_1 = -12.0616$
$\lambda_2 = -55.2717$

Por lo tanto, es sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada

→ Hipertenso

$z = 0.010$
$R = 1.000$
$L = 0.001$
$C = 1.000$

$a = 1\times10^{-5}$
$b = 1.01\times10^{-3}$
$C = 0.01$

$\lambda_1 = -11.1267$
$\lambda_2 = -89.8732$

Por lo tanto, el sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada

Error en estado estacionario

$$e(s) = \lim_{s \to 0} sPa(s)\left(1 - \frac{Pp(s)}{Pa(s)}\right) = C(s) \lim_{s \to 0} (s)\frac{1}{s}\left(1 - \frac{LRs + Rz}{CLRz s^2 + L(z+R)s + Rz}\right) = 0$$

$$e(t) = 0 \text{ v}$$

Modelo de ecuaciones integro-diferenciales

$$Fz(t) + FL(t) = Fc(t) + FR(t)$$

$$\frac{Pa(t)}{z} + \frac{1}{L}\int [Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt} = \frac{Pp(t)}{R} + \frac{Pp(t)}{z}$$

$$\left(\frac{1}{R} + \frac{1}{z}\right)Pp(t) + \frac{1}{L}\int [Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt}$$

$$Pp(t) = \left[\frac{Pa(t)}{z} + \frac{1}{L}\int [Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt}\right]\frac{Rz}{z+R}$$

Funciones de transferencias con valores

Hipotenso

$$\frac{Pp(s)}{Pa(s)} = \frac{1\times10^{-3} + 0.010}{3.5\times10^{-5} + 1.01\times10^{-3} + 0.010}$$

Normotenso

$$\frac{Pp(s)}{Pa(s)} = \frac{1\times10^{-3} + 0.010}{1.5\times10^{-5} + 1.01\times10^{-3} + 0.010}$$

Hipertenso

$$\frac{Pp(s)}{Pa(s)} = \frac{1\times10^{-3} + 0.010}{1\times10^{-5} + 1.01\times10^{-3} + 0.010}$$